

Gregory Li

(925) 353-8398 | gregory.li.ziyuan@gmail.com
linkedin.com/in/gregoryliziyuan | github.com/gregoryzli | gregoryzli.github.io

EDUCATION

University of California, Los Angeles — B.S. Computer Science, Expected June 2029 | GPA: 3.8/4.0

EXPERIENCE

UCLA ACM AI – AISF Neurosymbolic Research | Co-Researcher 2026

- Building a system that reduces LLM hallucination by storing model context in a Scallop-constrained knowledge graph, rejecting logically inconsistent state updates at write time.
- Implemented a recursive architecture segmenting long contexts into structured subchunks; benchmarking against GEPA and Dynamic Cheatsheet on LongBench, with a 4-person team preparing a NeurIPS workshop submission.

AVBotz Robotics Club | Software Developer 2021 – 2025

- Contributed to international RoboSub finishes of 2nd (2022), 5th (2023), and 14th (2024) against fields of 39, 34, and 41 teams.
- Migrated the vision pipeline from YOLOv5 to YOLOv8 after benchmarking against YOLOR on precision/recall, improving detection of small overlapping objects; built a Gazebo environment enabling subsystem validation without pool access.

Cybersecurity Club | Competitive Member 2023 – 2025

- Placed 28th nationally (top 2%) in PicoCTF 2025 among 1,817 U.S. teams with 5,610 points, solving binary exploitation, web, and cryptography challenges.

PROJECTS

SplitPay — Group Expense Splitter | React, TypeScript, Express, MySQL, Prisma | 2026

- Built and deployed a full-stack expense splitter with equal, exact, and percentage splits; JWT auth in an httpOnly cookie, group-membership checks on every scoped route, and two-party confirmation before a settlement affects balances.
- Implemented settle-up as a pure $O(n \log n)$ greedy matcher bounded by $n-1$ payments (exact minimum is NP-hard by reduction to subset-sum), with money as integer cents and largest-remainder allocation; 92 vitest/supertest tests including property-based checks.

gregoryzli.github.io/expense-splitter · github.com/gregoryzli/expense-splitter

Fuzz Testing — duckx C++ Library | Honggfuzz, llvm-cov | 2025

- Diagnosed why the fuzzer stalled: ZIP-structure validation rejected mutated inputs. Rewrote document.open to accept raw bytes and resolved pugixml link errors blocking instrumentation.
- Raised edge coverage $\sim 400 \rightarrow \sim 1,000$ across 237M executions; no memory-safety failures found.

Through the Gates of Hell — Enemy AI | Unity, C# | 2026

- Built enemy AI for a released top-down horror game: one script driving three archetypes (persistent tracker, vision-gated pursuer, delayed rusher) off a shared state machine with line-of-sight raycasting and per-instance tuning.
- Decoupled the AI from teammates' flashlight and player code behind two interfaces; light-based stun pins the destination each frame, freezing an in-progress path without forcing a replan.

github.com/gregoryzli/unity-astar-enemy-ai

GroupHub — Team Collaboration Web App | React, Express, SQLite | 2026

- Full-stack team management app with task tracking, messaging, role-based admin controls, JWT auth, and Playwright end-to-end tests across 109 commits with a 5-person team.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, JavaScript, C#, SQL

Frameworks: React, Next.js, Node.js, Express, Prisma, PyTorch, YOLO, Unity

Tools: Git, Linux, Docker, MySQL, SQLite, Gazebo, Ghidra, Burp Suite, Wireshark, vitest, Playwright